

MAC 2312 — CALCULUS II

Lecture 5: Integration by Parts (IBP)

Monday, September 14 • Sections 3.1

• Guided Notes •

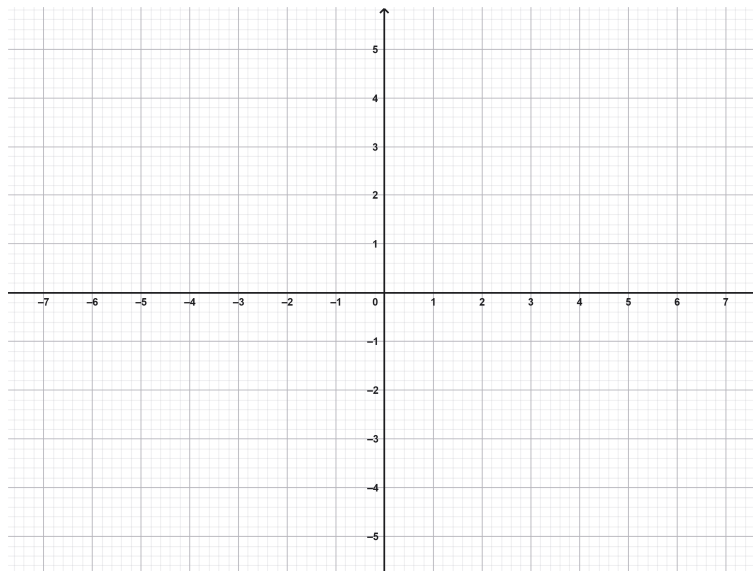
How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

We know already that integration can be used to calculate areas between curves, volumes of solids and arc lengths. But we don't have enough tools to evaluate the resulting integrals.

Example 5.1

Let $\mathcal{R}$ be the region bounded by $y = \ln(x) + 2$, the x -axis and the vertical lines $x = 1$ and $x = 4$. Set up the definite integrals equal to the volume of the solid that is obtained when $\mathcal{R}$ is rotated about the x -axis using the disks/washers and the shells methods.



Integration by parts is related to the product rule for differentiation in the same way that the substitution rule is related to the chain rule: it applies the product rule in reverse.

Theorem 5.1 – Integration by parts formula

For indefinite integrals: $\int u \, dv = uv - \int v \, du.$

For definite integrals: $\int_a^b u \, dv = uv|_a^b - \int_a^b v \, du.$

Example 5.2

Let us evaluate the integrals of the previous example using integration by parts.

Disks/Washers integral:

Shells integral:

What have we learned from the previous example?

1. Integration by parts can be used to evaluate both definite and indefinite integrals.
2. When the integrand consists of a single function,

$$\int f(x) \, dx,$$

a useful choice is often $u = f(x)$ and $dv = dx$.

3. When the integrand is a product of two or more functions, there may be several possible choices for u and dv .
4. A useful guideline for choosing u is LIATE:

Logarithmic > Inverse trigonometric > Algebraic > Trigonometric > Exponential

When possible, choose u from the function type appearing first in this list.

5. LIATE is a guideline, not a rule. A useful choice should make du simpler and should allow us to find v easily.
6. After applying integration by parts, we may need to apply it again, use substitution, apply another integration technique, or solve an equation involving the original integral.

Choosing u when applying integration by parts

Before applying integration by parts, ask:

1. Can the integrand be viewed as a product?
2. Which factor becomes simpler when differentiated?
3. Can I integrate the remaining factor to find v ?
4. Will the new integral be easier than the original integral?

Example 5.3

Let us evaluate the following integrals using integration by parts.

1. $\int \arcsin(x) \, dx.$

2. $\int_0^{\frac{\pi}{2}} x \sin(2x) \, dx.$

3. $\int x^2 e^{3x} \, dx.$

The examples above illustrate several important ideas. The first integral required both integration by parts and the substitution rule. In the second, we had several choices for u , but LIATE guided us to choose $u = x$ because Algebraic comes before Trigonometric. The third example illustrates both using LIATE to choose u and applying integration by parts more than once.

In some integrals, applying integration by parts produces another copy of the original integral. In this situation, we collect the original integrals on one side of the equation and solve for the integral.

Example 5.4

Let us determine $\int e^{2x} \cos(3x) \, dx.$

Let us practice integration by parts, first:

! Keep in mind

- Make sure that $u dv$ contains the entire integrand.
- Choose dv so that $v = \int dv$ can be found easily.
- Remember the minus sign in

$$\int u dv = uv - \int v du.$$

- Treat LIATE as a guideline, not a rule.
- Be prepared to apply integration by parts more than once.
- If the original integral reappears, solve algebraically for it.
- Include $+C$ only for indefinite integrals.
- For definite integrals, apply the bounds to the entire boundary term.
- Differentiate the final answer to check it when feasible.

Practice Problems

Level I problems

1. Determine

1. $\int x \cos(x) dx$.
2. $\int x \sec^2(x) dx$. Hint: $\int \sec^2(x) dx = \tan(x) + C$.
3. $\int x^3 \ln(x) dx$.
4. $\int x e^x dx$.
5. $\int x \ln(x^2) dx$. Hint: Before using integration by parts, check whether a substitution will work. When evaluating an integral, substitution should generally be one of the first techniques you consider.

2. Evaluate

1. $\int_1^4 \ln(\sqrt{x}) dx$.
2. $\int_1^5 \frac{\ln(x)}{x^2} dx$.
3. $\int_1^5 \frac{x}{e^x} dx$.

3. Find the volume of the solid obtained by rotating the region bounded by $y = \ln(x)$, $y = 0$, and $x = 3$; about the x -axis.

Practice Problems

Level II problems

1. Determine

1. $\int x \ln^2(x) dx$.
2. $\int x^2 \sin(2x) dx$.
3. $\int x^2 e^{-x} dx$.
4. $\int x^2 \tan^{-1}(x) dx$. Hint: Remember that $\tan^{-1}(x)$ is not $\frac{1}{\tan(x)}$, $\tan^{-1}(x) = \arctan(x)$.

2. Evaluate

1. $\int_1^2 t^2 e^t dt$.
2. $\int_0^1 x 3^x dx$.
3. $\int_0^{2\pi} t^2 \sin(2t) dt$.
4. $\int_1^{\sqrt{3}} \tan^{-1}\left(\frac{1}{x}\right) dx$.
5. $\int_0^1 (1+x^2) e^{3x} dx$.

3. Find the volume of the solid obtained by rotating the region bounded by $y = \ln(x)$, $y = 0$, and $x = 3$; about the x -axis.
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Practice Problems

Level III problems

Find the volume of the solid obtained by revolving the region bounded by the graph of $f(x) = \sin(x)$ and the x -axis over the interval $[0, \pi]$; about the line $y = -1$. Set up and evaluate the integrals that must be evaluated when the disk/washer and the shell methods are applied.

Checking Your Understanding

1. Identify **ALL** the statements which are true regarding the integral $I = \int x \ln(x) dx$.

1. Because $x \ln(x) = \frac{1}{2} 2x \ln(x) = \frac{1}{2} x \ln(x^2)$, after using the substitution $u = x^2$,

$$I = \frac{1}{4} \int \ln(u) du.$$

2. Using integration by parts with $u = \ln(x)$, $dv = xdx$,

$$I = \frac{1}{2} \left(x^2 \ln(x) - \int x dx \right).$$

3. Using integration by parts with $u = \ln(x)$, $dv = xdx$,

$$I = \frac{1}{2} \left(x^2 \ln(x) + \int x dx \right).$$

2. Identify **all** the statements which are true regarding the integral $I = \int \frac{\ln(x)}{\sqrt{x}} dx$.

1. Because $\frac{\ln(x)}{\sqrt{x}} = 2 \frac{1}{2} \frac{\ln(x)}{\sqrt{x}} = 2 \frac{\ln(\sqrt{x})}{\sqrt{x}}$, after using the substitution $u = \sqrt{x}$,

$$I = 4 \int \ln(u) du.$$

2. Using integration by parts with $u = \ln(x)$, $dv = \frac{dx}{\sqrt{x}}$,

$$I = 2\sqrt{x} \ln(x) - 2 \int \frac{1}{\sqrt{x}} dx.$$

3. Using integration by parts with $u = \ln(x)$, $dv = \frac{dx}{\sqrt{x}}$,

$$I = 2\sqrt{x} \ln(x) + 2 \int \frac{1}{\sqrt{x}} dx.$$

3. Suppose $f(x)$ is a differentiable function with antiderivative $F(x)$. From the given expressions, select **all** that are equal to $\int_a^b x f(x) dx$.

1. $\frac{1}{2} \left(x^2 f(x) \Big|_a^b + \int_a^b x^2 f'(x) dx \right).$

2. $xF(x) \Big|_a^b + \int_a^b F(x) dx.$

3. $xF(x) \Big|_a^b - \int_a^b F(x) dx.$

By the end of today you will be able to:

- Explain where the integration-by-parts formula comes from.
- Choose useful expressions for u and dv .
- Find du and v correctly.
- Evaluate definite and indefinite integrals using integration by parts.
- Apply integration by parts more than once when needed.
- Solve for the original integral when it reappears.
- Check an antiderivative by differentiation.